

OpenStax Calculus Volume 3

Section 2.4 Exercises: The Cross Product

Source: <https://openstax.org/books/calculus-volume-3/pages/2-4-the-cross-product>. Exercises reproduced from OpenStax, licensed under CC BY 4.0.

This document was converted programmatically from OpenStax content; formulas were translated automatically from MathML and may contain conversion errors — verify against the source before relying on them.

For the following exercises, the vectors $\mathbf{u}$ and $\mathbf{v}$ are given.

- (a) Find the cross product $\mathbf{u} \times \mathbf{v}$ of the vectors $\mathbf{u}$ and $\mathbf{v}$. Express the answer in component form.
- (b) Sketch the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$.

1. $\mathbf{u} = \langle 2, 0, 0 \rangle$, $\mathbf{v} = \langle 2, 2, 0 \rangle$
2. $\mathbf{u} = \langle 3, 2, -1 \rangle$, $\mathbf{v} = \langle 1, 1, 0 \rangle$
3. $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j}$, $\mathbf{v} = \mathbf{j} + 2\mathbf{k}$
4. $\mathbf{u} = 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{v} = 3\mathbf{i} + \mathbf{k}$
5. Simplify $(\mathbf{i} \times \mathbf{i} - 2\mathbf{i} \times \mathbf{j} - 4\mathbf{i} \times \mathbf{k} + 3\mathbf{j} \times \mathbf{k}) \times \mathbf{i}$.
6. Simplify $\mathbf{j} \times (\mathbf{k} \times \mathbf{j} + 2\mathbf{j} \times \mathbf{i} - 3\mathbf{j} \times \mathbf{j} + 5\mathbf{i} \times \mathbf{k})$.

In the following exercises, vectors $\mathbf{u}$ and $\mathbf{v}$ are given. Find unit vector $\mathbf{w}$ in the direction of the cross product vector $\mathbf{u} \times \mathbf{v}$. Express your answer using standard unit vectors.

7. $\mathbf{u} = \langle 3, -1, 2 \rangle$, $\mathbf{v} = \langle -2, 0, 1 \rangle$
8. $\mathbf{u} = \langle 2, 6, 1 \rangle$, $\mathbf{v} = \langle 3, 0, 1 \rangle$
9. $\mathbf{u} = \vec{AB}$, $\mathbf{v} = \vec{AC}$, where $A(1, 0, 1)$, $B(1, -1, 3)$, and $C(0, 0, 5)$
10. $\mathbf{u} = \vec{OP}$, $\mathbf{v} = \vec{PQ}$, where $P(-1, 1, 0)$ and $Q(0, 2, 1)$
11. Determine the real number α such that $\mathbf{u} \times \mathbf{v}$ and $\mathbf{i}$ are orthogonal, where $\mathbf{u} = 3\mathbf{i} + \mathbf{j} - 5\mathbf{k}$ and $\mathbf{v} = 4\mathbf{i} - 2\mathbf{j} + \alpha\mathbf{k}$.
12. Show that $\mathbf{u} \times \mathbf{v}$ and $2\mathbf{i} - 14\mathbf{j} + 2\mathbf{k}$ cannot be orthogonal for any α real number, where $\mathbf{u} = \mathbf{i} + 7\mathbf{j} - \mathbf{k}$ and $\mathbf{v} = \alpha\mathbf{i} + 5\mathbf{j} + \mathbf{k}$.
13. Show that $\mathbf{u} \times \mathbf{v}$ is orthogonal to $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$, where $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors.
14. Show that $\mathbf{v} \times \mathbf{u}$ is orthogonal to $(\mathbf{u} \cdot \mathbf{v})(\mathbf{u} + \mathbf{v}) + \mathbf{u}$, where $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors.
15. Calculate the determinant $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 7 \\ 2 & 0 & 3 \end{vmatrix}$.

16. Calculate the determinant $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & -4 \\ 1 & 6 & -1 \end{vmatrix}$.

For the following exercises, the vectors $\mathbf{u}$ and $\mathbf{v}$ are given. Use determinant notation to find vector $\mathbf{w}$ orthogonal to vectors $\mathbf{u}$ and $\mathbf{v}$.

17. $\mathbf{u} = \langle -1, 0, e^t \rangle$, $\mathbf{v} = \langle 1, e^{-t}, 0 \rangle$, where t is a real number

18. $\mathbf{u} = \langle 1, 0, x \rangle$, $\mathbf{v} = \langle \frac{2}{x}, 1, 0 \rangle$, where x is a nonzero real number

19. Find vector $(\mathbf{a} - 2\mathbf{b}) \times \mathbf{c}$, where $\mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 5 \\ 0 & 1 & 8 \end{vmatrix}$, $\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 1 \\ 2 & -1 & -2 \end{vmatrix}$, and $\mathbf{c} = \mathbf{i} + \mathbf{j} + \mathbf{k}$.

20. Find vector $\mathbf{c} \times (\mathbf{a} + 3\mathbf{b})$, where $\mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 0 & 9 \\ 0 & 1 & 0 \end{vmatrix}$, $\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 1 \\ 7 & 1 & -1 \end{vmatrix}$, and $\mathbf{c} = \mathbf{i} - \mathbf{k}$.

21. [T] Use the cross product $\mathbf{u} \times \mathbf{v}$ to find the acute angle between vectors $\mathbf{u}$ and $\mathbf{v}$, where $\mathbf{u} = \mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = \mathbf{i} + \mathbf{k}$. Express the answer in degrees rounded to the nearest integer.

22. [T] Use the cross product $\mathbf{u} \times \mathbf{v}$ to find the obtuse angle between vectors $\mathbf{u}$ and $\mathbf{v}$, where $\mathbf{u} = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = \mathbf{i} - 2\mathbf{j}$. Express the answer in degrees rounded to the nearest integer.

23. Use the sine and cosine of the angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ to prove Lagrange's identity: $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$.

24. Verify Lagrange's identity $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$ for vectors $\mathbf{u} = -\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$.

25. Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are called collinear if there exists a nonzero scalar α such that $\mathbf{v} = \alpha\mathbf{u}$. Show that $\mathbf{u}$ and $\mathbf{v}$ are collinear if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.

26. Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are called collinear if there exists a nonzero scalar α such that $\mathbf{v} = \alpha\mathbf{u}$. Show that vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are collinear, where $A(4, 1, 0)$, $B(6, 5, -2)$, and $C(5, 3, -1)$.

27. Find the area of the parallelogram with adjacent sides $\mathbf{u} = \langle 3, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 2, 1 \rangle$.

28. Find the area of the parallelogram with adjacent sides $\mathbf{u} = \mathbf{i} + \mathbf{j}$ and $\mathbf{v} = \mathbf{i} + \mathbf{k}$.

29. Consider points $A(3, -1, 2)$, $B(2, 1, 5)$, and $C(1, -2, -2)$.

(a) Find the area of parallelogram $ABCD$ with adjacent sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

(b) Find the area of triangle ABC .

(c) Find the distance from point A to line BC .

30. Consider points $A(2, -3, 4)$, $B(0, 1, 2)$, and $C(-1, 2, 0)$.

(a) Find the area of parallelogram $ABCD$ with adjacent sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

(b) Find the area of triangle ABC .

(c) Find the distance from point B to line AC .

In the following exercises, vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are given.

(a) Find the triple scalar product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.

(b) Find the volume of the parallelepiped with the adjacent edges $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

31. $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j} + \mathbf{k}$, and $\mathbf{w} = \mathbf{i} + \mathbf{k}$

32. $\mathbf{u} = \langle -3, 5, -1 \rangle$, $\mathbf{v} = \langle 0, 2, -2 \rangle$, and $\mathbf{w} = \langle 3, 1, 1 \rangle$

33. Calculate the triple scalar products $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{w})$ and $\mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})$, where $\mathbf{u} = \langle 1, 1, 1 \rangle$, $\mathbf{v} = \langle 7, 6, 9 \rangle$, and $\mathbf{w} = \langle 4, 2, 7 \rangle$.

34. Calculate the triple scalar products $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{u})$ and $\mathbf{u} \cdot (\mathbf{w} \times \mathbf{v})$, where $\mathbf{u} = \langle 4, 2, -1 \rangle$, $\mathbf{v} = \langle 2, 5, -3 \rangle$, and $\mathbf{w} = \langle 9, 5, -10 \rangle$.

35. Find vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ with a triple scalar product given by the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 2 & 5 \\ 8 & 9 & 2 \end{vmatrix}$.

Determine their triple scalar product.

36. The triple scalar product of vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is given by the determinant $\begin{vmatrix} 0 & -2 & 1 \\ 0 & 1 & 4 \\ 1 & -3 & 7 \end{vmatrix}$. Find vector $\mathbf{a} - \mathbf{b} + \mathbf{c}$.

37. Consider the parallelepiped with edges OA , OB , and OC , where $A(2, 1, 0)$, $B(1, 2, 0)$, and $C(0, 1, \alpha)$.

(a) Find the real number $\alpha > 0$ such that the volume of the parallelepiped is 3 units³.

(b) For $\alpha = 1$, find the height h from vertex C of the parallelepiped to the plane formed by the edges OA and OB .

38. Consider points $A(\alpha, 0, 0)$, $B(0, \beta, 0)$, and $C(0, 0, \gamma)$, with α , β , and γ positive real numbers.

(a) Determine the volume of the parallelepiped with adjacent sides $\vec{OA}$, $\vec{OB}$, and $\vec{OC}$.

(b) Find the volume of the tetrahedron with vertices O , A , B , and C . (Hint: The volume of the tetrahedron is 1/6 of the volume of the parallelepiped.)

(c) Find the distance from the origin to the plane determined by A , B , and C . Sketch the parallelepiped and tetrahedron.

39. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be three-dimensional vectors and c be a real number. Prove the following properties of the cross product.

(a) $\mathbf{u} \times \mathbf{u} = \mathbf{0}$

(b) $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$

(c) $c(\mathbf{u} \times \mathbf{v}) = (c\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (c\mathbf{v})$

(d) $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$

40. Show that vectors $\mathbf{u} = \langle 1, 0, -8 \rangle$, $\mathbf{v} = \langle 0, 1, 6 \rangle$, and $\mathbf{w} = \langle -1, 9, 3 \rangle$ satisfy the following properties of the cross product.

(a) $\mathbf{u} \times \mathbf{u} = \mathbf{0}$

- (b) $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$
- (c) $c(\mathbf{u} \times \mathbf{v}) = (c\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (c\mathbf{v})$
- (d) $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$
41. Nonzero vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are said to be linearly dependent if one of the vectors is a linear combination of the other two. For instance, there exist two nonzero real numbers α and β such that $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$. Otherwise, the vectors are called linearly independent. Show that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are coplanar if and only if they are linearly dependent.
42. Consider vectors $\mathbf{u} = \langle 1, 4, -7 \rangle$, $\mathbf{v} = \langle 2, -1, 4 \rangle$, $\mathbf{w} = \langle 0, -9, 18 \rangle$, and $\mathbf{p} = \langle 0, -9, 17 \rangle$.
- (a) Show that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are coplanar by using their triple scalar product
- (b) Show that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are coplanar, using the definition that there exist two nonzero real numbers α and β such that $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$.
- (c) Show that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{p}$ are linearly independent—that is, none of the vectors is a linear combination of the other two.
43. Consider points $A(0, 0, 2)$, $B(1, 0, 2)$, $C(1, 1, 2)$, and $D(0, 1, 2)$. Are vectors $\vec{AB}$, $\vec{AC}$, and $\vec{AD}$ linearly dependent (that is, one of the vectors is a linear combination of the other two)?
44. Show that vectors $\mathbf{i} + \mathbf{j}$, $\mathbf{i} - \mathbf{j}$, and $\mathbf{i} + \mathbf{j} + \mathbf{k}$ are linearly independent—that is, there do not exist two nonzero real numbers α and β such that $\mathbf{i} + \mathbf{j} + \mathbf{k} = \alpha(\mathbf{i} + \mathbf{j}) + \beta(\mathbf{i} - \mathbf{j})$.
45. Let $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$ be two-dimensional vectors. The cross product of vectors $\mathbf{u}$ and $\mathbf{v}$ is not defined. However, if the vectors are regarded as the three-dimensional vectors $\tilde{\mathbf{u}} = \langle u_1, u_2, 0 \rangle$ and $\tilde{\mathbf{v}} = \langle v_1, v_2, 0 \rangle$, respectively, then, in this case, we can define the cross product of $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{v}}$. In particular, in determinant notation, the cross product of $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{v}}$ is given by
- $$\tilde{\mathbf{u}} \times \tilde{\mathbf{v}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & 0 \\ v_1 & v_2 & 0 \end{vmatrix}.$$
- Use this result to compute $(\mathbf{i} \cos \theta + \mathbf{j} \sin \theta) \times (\mathbf{i} \sin \theta - \mathbf{j} \cos \theta)$, where θ is a real number.
46. Consider points $P(2, 1)$, $Q(4, 2)$, and $R(1, 2)$.
- (a) Find the area of triangle P , Q , and R .
- (b) Determine the distance from point R to the line passing through P and Q .
47. Determine a vector of magnitude 10 perpendicular to the plane passing through the x-axis and point $P(1, 2, 4)$.
48. Determine a unit vector perpendicular to the plane passing through the z-axis and point $A(3, 1, -2)$.
49. Consider $\mathbf{u}$ and $\mathbf{v}$ two three-dimensional vectors. If the magnitude of the cross product vector $\mathbf{u} \times \mathbf{v}$ is k times larger than the magnitude of vector $\mathbf{u}$, show that the magnitude of $\mathbf{v}$ is greater than or equal to k , where k is a natural number.
50. [T] Assume that the magnitudes of two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are known. The function $f(\theta) = \|\mathbf{u}\|\|\mathbf{v}\|\sin \theta$ defines the magnitude of the cross product vector $\mathbf{u} \times \mathbf{v}$, where $\theta \in [0, \pi]$ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

- (a) Graph the function f .
- (b) Find the absolute minimum and maximum of function f . Interpret the results.
- (c) If $\|\mathbf{u}\| = 5$ and $\|\mathbf{v}\| = 2$, find the angle between $\mathbf{u}$ and $\mathbf{v}$ if the magnitude of their cross product vector is equal to 9.
51. Find all vectors $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$ that satisfy the equation $\langle 1, 1, 1 \rangle \times \mathbf{w} = \langle -1, -1, 2 \rangle$.
52. Solve the equation $\mathbf{w} \times \langle 1, 0, -1 \rangle = \langle 3, 0, 3 \rangle$, where $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$ is a nonzero vector with a magnitude of 3.
53. [T] A mechanic uses a 12-in. wrench to turn a bolt. The wrench makes a 30° angle with the horizontal. If the mechanic applies a vertical force of 10 lb on the wrench handle, what is the magnitude of the torque at point P (see the following figure)? Express the answer in foot-pounds rounded to two decimal places.
[Figure: This figure is the image of an open-end wrench. The lower portion of the wrench is at point P . The wrench has a length of "12 in." The angle the wrench makes with a horizontal line from P is 30 degrees. At the top of the wrench is a downward vertical vector labeled "10 lb".]
54. [T] A boy applies the brakes on a bicycle by applying a downward force of 20 lb on the pedal when the 6-in. crank makes a 40° angle with the horizontal (see the following figure). Find the torque at point P . Express your answer in foot-pounds rounded to two decimal places.
[Figure: This figure shows the pedals, cranks, and chain of a bicycle. The distance along the crank to the top pedal is 6 in. The angle of the crank is 40 degrees with the horizontal, measured toward the rear. The top pedal has a downward vector labeled "20 lb".]
55. [T] Find the magnitude of the force that needs to be applied to the end of a 20-cm wrench located on the positive direction of the y -axis if the force is applied in the direction $\langle 0, 1, -2 \rangle$ and it produces a 100 N·m torque to the bolt located at the origin.
56. [T] What is the magnitude of the force required to be applied to the end of a 1-ft wrench at an angle of 35° to produce a torque of 20 ft-lbs?
57. [T] The force vector $\mathbf{F}$ acting on a proton with an electric charge of $1.6 \times 10^{-19}\text{C}$ (in coulombs) moving in a magnetic field $\mathbf{B}$ where the velocity vector $\mathbf{v}$ is given by $\mathbf{F} = 1.6 \times 10^{-19}(\mathbf{v} \times \mathbf{B})$ (here, $\mathbf{v}$ is expressed in meters per second, $\mathbf{B}$ is in tesla [T], and $\mathbf{F}$ is in newtons [N]). Find the force that acts on a proton that moves in the xy -plane at velocity $\mathbf{v} = 10^5\mathbf{i} + 10^5\mathbf{j}$ (in meters per second) in a magnetic field given by $\mathbf{B} = 0.3\mathbf{j}$.
58. [T] The force vector $\mathbf{F}$ acting on a proton with an electric charge of $1.6 \times 10^{-19}\text{C}$ moving in a magnetic field $\mathbf{B}$ where the velocity vector $\mathbf{v}$ is given by $\mathbf{F} = 1.6 \times 10^{-19}(\mathbf{v} \times \mathbf{B})$ (here, $\mathbf{v}$ is expressed in meters per second, $\mathbf{B}$ in T, and $\mathbf{F}$ in N). If the magnitude of force $\mathbf{F}$ acting on a proton is $5.9 \times 10^{-17}\text{ N}$ and the proton is moving at the speed of 300 m/sec in magnetic field $\mathbf{B}$ of magnitude 2.4 T, find the angle between velocity vector $\mathbf{v}$ of the proton and magnetic field $\mathbf{B}$. Express the answer in degrees rounded to the nearest integer.
59. [T] Consider $\mathbf{r}(t) = \langle \cos t, \sin t, 2t \rangle$ the position vector of a particle at time $t \in [0, 30]$, where the components of $\mathbf{r}$ are expressed in centimeters and time in seconds. Let $\vec{OP}$ be the position vector of the particle after 1 sec.
- (a) Determine unit vector $\mathbf{B}(t)$ (called the binormal unit vector) that has the direction of cross product vector $\mathbf{v}(t) \times \mathbf{a}(t)$, where $\mathbf{v}(t)$ and $\mathbf{a}(t)$ are the instantaneous velocity vector and, respectively, the acceleration vector of the particle after t seconds.

- (b) Use a CAS to visualize vectors $\mathbf{v}(1)$, $\mathbf{a}(1)$, and $\mathbf{B}(1)$ as vectors starting at point P along with the path of the particle.
60. A solar panel is mounted on the roof of a house. The panel may be regarded as positioned at the points of coordinates (in meters) $A(8, 0, 0)$, $B(8, 18, 0)$, $C(0, 18, 8)$, and $D(0, 0, 8)$ (see the following figure).
[Figure: This figure shows a rectangular set of solar panels on a roof. The corners are labeled "A, B, C, D." Also there is a vector drawn from A to D. There is another vector along the bottom of the rectangle from A to B.]
- (a) Find vector $\mathbf{n} = \vec{AB} \times \vec{AD}$ perpendicular to the surface of the solar panels. Express the answer using standard unit vectors.
- (b) Assume unit vector $\mathbf{s} = \frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$ points toward the Sun at a particular time of the day and the flow of solar energy is $\mathbf{F} = 900\mathbf{s}$ (in watts per square meter $[\text{W}/\text{m}^2]$). Find the predicted amount of electrical power the panel can produce, which is given by the dot product of vectors $\mathbf{F}$ and $\mathbf{n}$ (expressed in watts).
- (c) Determine the angle of elevation of the Sun above the solar panel. Express the answer in degrees rounded to the nearest whole number. (Hint: The angle between vectors $\mathbf{n}$ and $\mathbf{s}$ and the angle of elevation are complementary.)